OM THAKUR

Senior Software Engineer / SDE-2 | Java Backend Architect

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PROFESSIONAL SUMMARY

Accomplished Senior Software Engineer / SDE-2 with 3.5 years of experience (3 years full-time + 6 months internship) designing, architecting, and optimizing high-throughput distributed backend ecosystems. Deep specialization in Java 21, Spring Boot 3, Apache Kafka streaming, distributed cache fabrics, database sharding, and cloud-native Kubernetes environments. Proven record of owning Tier-1 services processing 1M+ daily transactions, driving 99.99% service reliability, and cutting cloud infrastructure expenditures by 32%. Passionate technical leader, system architect, and engineering mentor.

TECHNICAL SKILLS

Languages: Java (17/21), SQL, Go, Python, Bash
Enterprise Frameworks: Spring Boot 3, Spring Cloud (Gateway, Eureka, Config), Hibernate / JPA, Micrometer, Resilience4j
Distributed Architecture: Event-Driven Microservices, Apache Kafka (Streams, Connect, Schema Registry), RabbitMQ, Distributed Caching (Redis Cluster), CQRS, Event Sourcing
Data Stores & Search: MySQL, PostgreSQL, MongoDB, Elasticsearch, Redis, Cassandra (basics), Flyway
Cloud & DevOps: Kubernetes (K8s), Docker, Helm, AWS (EKS, MSK, RDS, S3, CloudFront), Terraform, GitHub Actions, ArgoCD
Observability & Reliability: Prometheus, Grafana, OpenTelemetry, Jaeger, ELK Stack, Distributed Tracing, Chaos Engineering
Testing & Methodologies: TDD, JUnit 5, Mockito, Testcontainers, Domain-Driven Design (DDD), Agile/Scrum Leadership

PROFESSIONAL EXPERIENCE

Senior Software Engineer (Backend) | Apex Fintech Solutions (or Enterprise SaaS), Indore / Remote *October 2025 - Present (1 Year)*

- Spearheaded architectural evolution of Tier-1 transactional payment and ledger services, scaling throughput from 2,000 to 12,000 TPS while maintaining P99 latency below 80ms.
- Architected event-driven distributed streaming pipelines using Apache Kafka and Kafka Streams, handling 2.5M+ events daily with zero data loss guarantees.
- Instituted multi-region Redis cluster caching layer with automated write-behind synchronization, decreasing primary database query pressure by 60%.
- Authored Terraform IaC configurations and ArgoCD GitOps pipelines for automated multi-environment Kubernetes deployments on AWS EKS.
- Championed engineering excellence: conducted high-level design (HLD/LLD) reviews, established team-wide code standards, and mentored 6 engineers.

Software Development Engineer (SDE-1) | Tech Corp, Indore *October 2024 - September 2025 (1 Year)*

- Engineered 25+ resilient microservices endpoints using Java 17 and Spring Boot 3, serving 150,000+ daily active users.
- Re-architected slow database queries and executed table partitioning across PostgreSQL tables, cutting P95 response times by 65%.
- Integrated STOMP WebSockets over SockJS cluster for real-time order state updates and user notifications.
- Built end-to-end integration test suites utilizing Testcontainers and Mockito, increasing test coverage from 55% to 84%.

Associate Software Engineer | Dollop Infotech, Indore *October 2023 - September 2024 (1 Year)*

- Developed backend business modules using Spring Boot, Spring Data JPA, and MySQL, implementing clean DTO and repository patterns.
- Configured stateless JWT security layers with Spring Security and BCrypt encryption, preventing CSRF and unauthorized API access.

Software Engineer Intern | Dollop Infotech, Indore *April 2023 - September 2023 (6 Months Internship)*

- Designed and built 15+ secure RESTful APIs for enterprise web platforms using Java, Spring Boot, and Spring Data JPA.
- Configured Spring Security with stateless JWT authentication, password encryption (BCrypt), and role-based access control.
- Optimized relational database access in MySQL, resolving Hibernate N+1 query bottlenecks and writing automated unit test suites with JUnit 5 and Mockito.

KEY TECHNICAL PROJECTS

High-Throughput Distributed Social & Feed Engine (Instagram Scale Architecture) | *GitHub: github.com/Omsingh1912/Instagram-clone*

Tech: Java 21, Spring Boot 3, Apache Kafka, Redis Cluster, WebSockets, PostgreSQL, Docker, AWS EKS

- Architected a distributed social media platform capable of handling real-time post creation, feed fan-out, multimedia indexing, and instant messaging.
- Implemented a hybrid feed generation mechanism (fan-out-on-write for standard users, fan-out-on-read for high-follower accounts) backed by Redis sorted sets (ZSET), slashing feed generation latency by 80%.
- Built real-time notifications and live interaction streams using Spring WebSocket (STOMP) cluster backed by Redis Pub/Sub for cross-pod synchronization.
- Containerized services with Docker and deployed via Helm charts on Kubernetes with automated horizontal pod autoscaling (HPA) driven by CPU and custom request metrics.

Enterprise Multi-Tenant Sales Pipeline & Analytics Platform

Tech: Java 17, Spring Boot, Spring Security, Redis, MySQL, Flyway, Docker Compose

- Designed a multi-tenant sales CRM supporting dynamic lead scoring, deal velocity analytics, and sales agent performance dashboards.
- Implemented asynchronous batch processing for multi-million record CSV ingestion, utilizing multi-threaded batch readers and direct JDBC batch inserts.
- Enforced automated schema migrations across all tenant databases using Flyway, ensuring strict schema consistency.

PROBLEM SOLVING & ALGORITHMS

250+ LeetCode problems solved. Advanced system design expertise: Microservices Decomposition, CAP Theorem, Database Sharding, Eventual Consistency, Idempotency, Rate Limiting Algorithms.

EDUCATION

Bachelor of Technology (B.Tech) - Information Technology | Sushila Devi Bansal College of Technology (SDBCT), Indore *Graduation (SD 2023) | CGPA: 6.9 / 10*